MAJOR PROJECT REPORT

Heart Disease Prediction Using Machine Learning Classification Models

A comparison of four supervised classification approaches

PROJECT FOCUS	DATASET
Binary classification, model comparison, feature interpretation, and an optional prediction demo.	303 supplied rows; 302 unique rows used; 13 input features and a binary target.

Prepared from the supplied project brief and dataset

25 October 2026

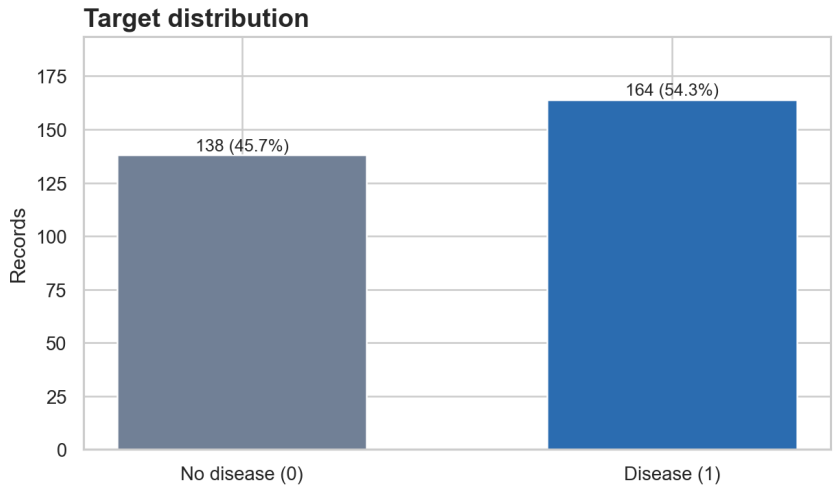
Abstract

This project compares Logistic Regression, Decision Tree, Random Forest, and a small feedforward neural network for predicting the supplied binary heart-disease target from 13 patient attributes. The worksheet contains 303 records and 14 columns. It has no missing cells and one exact duplicate row; after removing that row, 302 records were used. Hyperparameters were tuned with five-fold stratified cross-validation on the training partition. Logistic Regression had the strongest mean cross-validation ROC-AUC (0.911) and was selected. On the held-out test set it achieved 0.820 accuracy, 0.844 precision, 0.818 recall, 0.831 F1-score, and 0.893 ROC-AUC. The Random Forest had the highest test ROC-AUC (0.904), but its cross-validation score was slightly lower. Results on this small sample are educational and do not establish clinical performance.

Dataset summary after duplicate removal

Item	Value
Rows used for analysis	302 (303 supplied; one exact duplicate removed)
Input features / target	13 input features / binary target
Target 0	138 (45.7%)
Target 1	164 (54.3%)
Missing cells	0
Original file format	Excel workbook saved with a .csv filename; exported to a proper CSV for this package

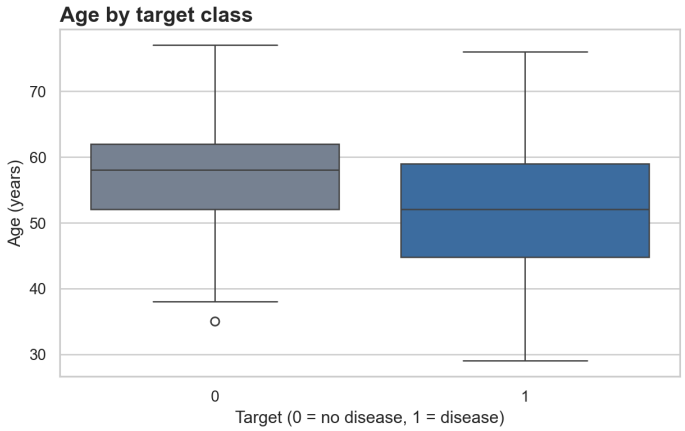
Target class counts



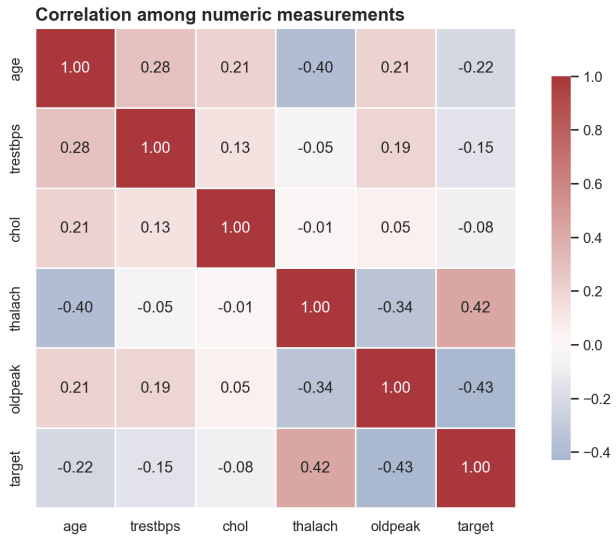
Exploratory analysis and data preparation

The two target classes are represented at similar rates in the supplied records. The age distributions overlap. Among the five continuous measurements, the largest simple correlations with the binary target are 0.42 for maximum heart rate (`thalach`) and -0.43 for `oldpeak`; these are descriptive relationships and not causal findings. The correlation chart includes continuous measurements only, because the dataset's category codes are not treated as a numeric scale.

Age by target class



Numeric measurement correlations



Five numeric columns were median-imputed and standardized. Eight coded columns (`sex`, `cp`, `fbs`, `restecg`, `exang`, `slope`, `ca`, `thal`) were most-frequent imputed and one-hot encoded. There were no missing cells, but imputation is retained inside the pipelines for future input files. All supplied code values were kept; the workbook does not include a complete verified codebook.

Modeling methodology

A stratified 80/20 split produced 241 training rows and 61 test rows (`random_state=42`). Each model uses the same preprocessing pipeline. Grid search with five-fold stratified cross-validation selected hyperparameters using mean ROC-AUC on training data only. The final model family was selected by the highest training cross-validation ROC-AUC; the test partition was held out for performance comparison. Precision, recall, and F1 use target 1 as the positive class at the 0.50 probability threshold.

Model	Tuned settings
Logistic Regression	C = 0.1, 1, 10; class weight = none or balanced
Decision Tree	Depth = 3 or unlimited; leaf size = 2 or 5; class weight = none or balanced
Random Forest	300 trees; depth = 6 or unlimited; leaf size = 1 or 3; class weight = none or balanced
Feedforward Neural Network	Hidden layers = (16), (32), or (32, 16); alpha = 0.0001 or 0.01

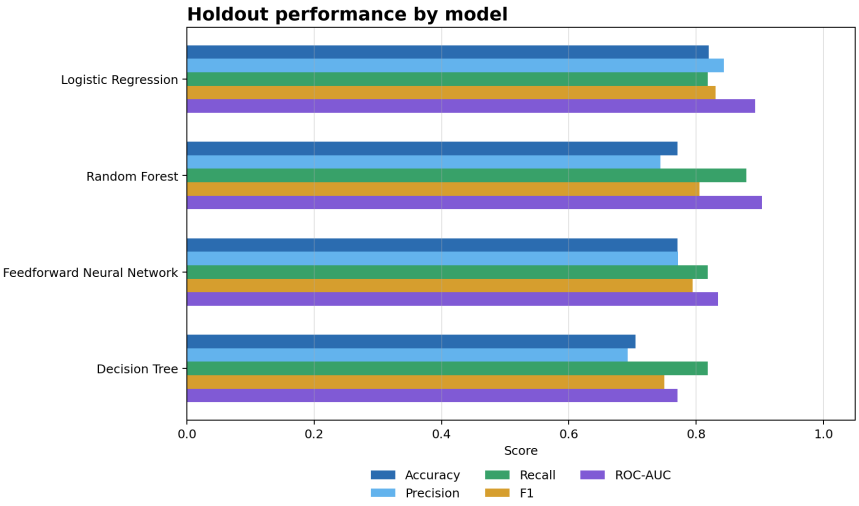
Held-out test results

Model	CV AUC	Accuracy	Precision	Recall	F1	Test AUC
Logistic Regression	0.911	0.820	0.844	0.818	0.831	0.893
Random Forest	0.909	0.770	0.744	0.879	0.806	0.904
Feedforward Neural Network	0.874	0.770	0.771	0.818	0.794	0.834
Decision Tree	0.847	0.705	0.692	0.818	0.750	0.771

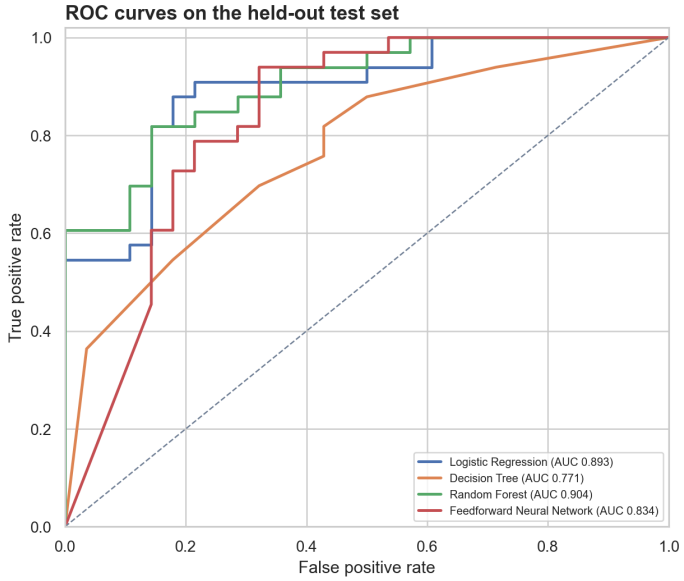
Logistic Regression was selected because it had the highest mean cross-validation ROC-AUC (0.911). Random Forest performed slightly better on the single test ROC-AUC. Since the scores are close and the test set is small, this difference is not enough to establish a reliable winner.

Evaluation visuals

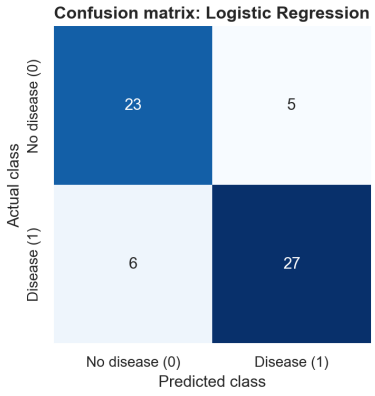
All four tuned models were evaluated on the same held-out test records.



ROC curves on the held-out set

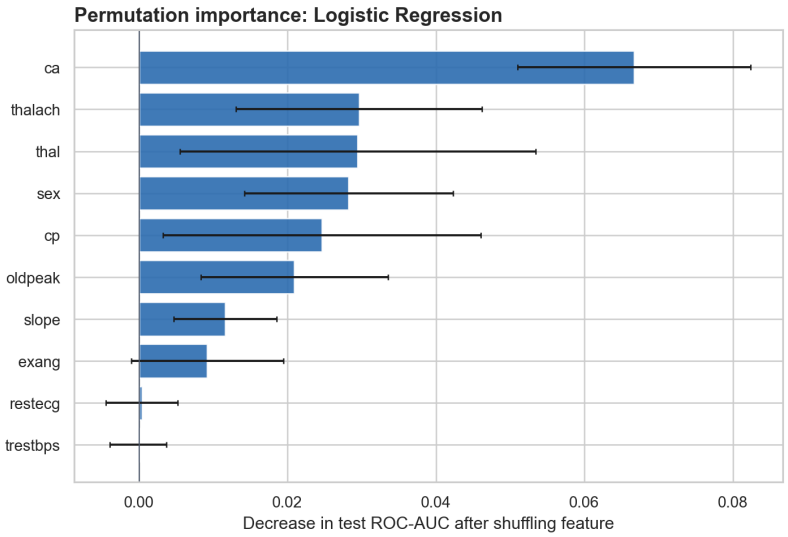


Selected model confusion matrix (actual rows x predicted columns): TN=23, FP=5, FN=6, TP=27. At the chosen threshold, the model correctly classified 23 target-0 and 27 target-1 records.



Feature interpretation

Permutation importance was calculated by shuffling one feature at a time on the held-out test data and measuring the change in ROC-AUC. The largest mean decreases were: ca (0.067), thalach (0.030), thal (0.029), sex (0.028), cp (0.025). These values describe predictive contribution for this model and split; they do not mean that a feature causes disease. Near-zero or negative scores can occur on a small test set, especially when predictors contain overlapping information.



Conclusion

The project completed the requested data review, exploratory analysis, model tuning, evaluation, and feature-importance steps. Logistic Regression had the strongest mean cross-validation score and reached 0.893 test ROC-AUC. Random Forest reached 0.904 test ROC-AUC, with a slightly lower cross-validation score. Repeated or nested cross-validation and validation on independent data would be needed for more stable estimates. The package also includes a Streamlit interface that loads the selected model and demonstrates a class-1 probability output.

Limitations

- The analysis uses 302 unique records from one supplied dataset; the test set has 61 records.
- A single holdout can produce variable estimates, and the exact source provenance and code mappings are not confirmed.
- Performance on these records does not establish generalization to other populations or clinical settings.
- Calibration, clinical utility, fairness, and the consequences of errors were not assessed.
- The model is an educational demonstration, not a diagnostic or treatment tool.

References

1. User-supplied project brief, *Heart Disease Prediction Using Machine Learning Classification Models*.

2. UCI Machine Learning Repository. (n.d.). *Heart Disease*. https://archive.ics.uci.edu/dataset/45/heart+disease. The supplied workbook's exact provenance is not confirmed.

3. Scikit-learn developers. (n.d.). *GridSearchCV*. scikit-learn.org/stable/modules/generated/sklearn.model_selection.GridSearchCV.html.

4. Scikit-learn developers. (n.d.). *MLPClassifier*. scikit-learn.org/stable/modules/generated/sklearn.neural_network.MLPClassifier.html.

Educational scope: The model's probability output describes how this fitted classifier responds to values similar to the supplied dataset. It is not an individual's clinical risk estimate and must not be used for diagnosis, screening, or treatment.